

GENERAL

1. ALL ACTIVITIES, WORKMANSHIP AND MATERIALS ARE TO BE IN ACCORDANCE WITH THE RELEVANT PROVISIONS OF THE QUEENSLAND BUILDING ACT, THE RELEVANT PROVISIONS OF THE NATIONAL CONSTRUCTION CODE, THE RELEVANT PROVISIONS OF THE RELEVANT AUSTRALIAN STANDARDS, THE RELEVANT PROVISIONS OF THE CAPRICORN MUNICIPAL DEVELOPMENT GUIDELINES, THESE CIVIL ENGINEERING DRAWINGS AND THE PROJECT SPECIFICATIONS.
2. THE CONTRACTOR IS TO COMPLY WITH THE REQUIREMENTS OF THE LOCAL AUTHORITY AND THE BUILDING CERTIFIER. ENSURE ALL PERMITS ARE IN PLACE BEFORE CONSTRUCTION.
3. ALL CONTRACTORS AND SUB-CONTRACTORS ARE TO HOLD APPROPRIATE REGISTRATION AND LICENSES.
4. PERFORM ALL ACTIVITIES IN A SAFE AND RESPONSIBLE MANNER AND IN ACCORDANCE WITH THE RELEVANT PROVISIONS OF THE QUEENSLAND WORK, HEALTH AND SAFETY ACT, QUEENSLAND WORK HEALTH AND SAFETY REGULATION, CODES OF PRACTICE AND INDUSTRY ACCEPTED CONSTRUCTION PRACTICE AND PRINCIPLES.
5. DURING THE CIVIL ENGINEERING DESIGN PROCESS FOR THIS PROJECT THE SAFETY OF THE PEOPLE WHO WILL CONSTRUCT, USE, MAINTAIN AND DEMOLISH THE CIVIL ELEMENTS HAVE BEEN CONSIDERED. RISKS TO THESE PEOPLE HAVE BEEN IDENTIFIED AND ARE DOCUMENTED IN THE SAFETY IN DESIGN REPORT.
6. ALL EXISTING SERVICES SHOWN ARE APPROXIMATE ONLY.

6.1. EXISTING SERVICES ARE SHOWN APPROXIMATELY BASED OFF RECORDS. THE CONTRACTOR IS TO CONFIRM LOCATION AND DEPTH OF ALL EXISTING SERVICES PRIOR TO CONSTRUCTION.

6.2. WHILST CARE HAS BEEN TAKEN TO IDENTIFY ALL SERVICES, SOME SERVICES MAY BE UNIDENTIFIED AND THE CONTRACTOR IS TO PERFORM THEIR OWN INVESTIGATION TO SATISFY THEMSELVES OF THE LOCATION AND DEPTH OF SERVICES PRIOR TO CONSTRUCTION.

6.3. FOR EXTENTS OF DEMOLITION (INCLUDING REMOVAL AND REALIGNING OF EXISTING SERVICES) REFER TO ARCHITECTURAL DOCUMENTATION.
7. ALL EXISTING SERVICES ARE TO BE LOCATED AND PROTECTED SO AS TO PREVENT DAMAGE.
8. DO NOT SCALE, PHYSICALLY OR ELECTRONICALLY, FROM THESE CIVIL ENGINEERING DRAWINGS.
9. THESE CIVIL ENGINEERING DRAWINGS HAVE BEEN BASED ON UNVERIFIED EXISTING DRAWINGS AND LIMITED VISUAL INSPECTIONS OF THE EXISTING CIVIL ELEMENTS. THE CONTRACTOR IS TO EXPOSE THE EXISTING CIVIL ELEMENTS TO ALLOW THE CIVIL ENGINEER TO CONFIRM THESE CIVIL ENGINEERING DRAWINGS BEFORE CONSTRUCTION AND FABRICATION BEGINS.
10. READ THESE CIVIL ENGINEERING DRAWINGS IN CONJUNCTION WITH ALL ARCHITECTURAL DRAWINGS, OTHER CONSULTANTS' DRAWINGS, THE PROJECT SPECIFICATIONS AND OTHER WRITTEN INSTRUCTIONS THAT MAY BE ISSUED. REFER ANY DISCREPANCIES TO THE CIVIL ENGINEER FOR CLARIFICATION BEFORE PROCEEDING WITH THAT PART OF WORKS.
11. UNLESS NOTED OTHERWISE , ALL STRUCTURAL DIMENSIONS ARE IN MILLIMETRES, ALL CIVIL DIMENSIONS ARE IN METRES AND ALL LEVELS ARE IN METRES TO THE AUSTRALIAN HEIGHT DATUM.
12. ALL PENETRATIONS, CHASES ETC IN CIVIL ELEMENTS ARE TO BE APPROVED BY THE CIVIL ENGINEER BEFORE CONSTRUCTION UNLESS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS.
13. INSTALL ALL PROPRIETARY PRODUCTS IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATIONS, RECOMMENDATIONS AND DETAILS.
14. ANY DEVIATION OR SUBSTITUTION FROM THE INFORMATION IN THESE CIVIL ENGINEERING DRAWINGS IS TO BE APPROVED BY THE CIVIL ENGINEER BEFORE CONSTRUCTION FABRICATION BEGINS.
15. DURING CONSTRUCTION MAINTAIN THE WORKS ARE IN A STABLE CONDITION AND ENSURE NO PART IS OVERSTRESSED. PROVIDE ALL TEMPORARY PROPPING, BRACING AND SHORING NECESSARY TO ACHIEVE THIS. ANY TEMPORARY WORKS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS ARE INDICATIVE ONLY.
16. WHERE CIVIL ENGINEERING CERTIFICATION OF CONSTRUCTION IS REQUIRED, THE CIVIL ENGINEER IS TO PERFORM ADEQUATE SITE VISITS TO ENSURE THE DESIGN INTENT OF THESE CIVIL ENGINEERING DRAWINGS IS ACHIEVED DURING CONSTRUCTION. THE CONTRACTOR IS TO GIVE A MINIMUM OF 48 HOURS NOTICE BEFORE THE COMMENCEMENT AND BEFORE THE COMPLETION OF ALL CIVIL ELEMENTS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS. THE CIVIL ENGINEER REQUIRES INSPECTIONS AT THE FOLLOWING STAGES OF CONSTRUCTION.

16.1. AT SUBGRADE LEVEL OF ALL PAVEMENT BOXES.

16.2. AT SUBBASE LEVEL PRIOR TO CONCRETE FORMWORK SETUP.

16.3. SETUP AND REINFORCING STEEL BEFORE CONCRETE PLACEMENT.

16.4. AT COMPLETION OF CIVIL WORKS.
17. THE CONTRACTOR IS TO UNDERTAKE LEVEL VERIFICATION CHECKS OF PAVEMENT LAYERS AND PROVIDE THIS INFORMATION TO THE ENGINEER FOR REVIEW UPON REQUEST.
18. QUALITY ASSURANCE TESTING IS TO BE PROVIDED AS PER THE REQUIREMENTS STATED ON THESE CIVIL ENGINEERING DRAWINGS.
19. THE DETAILED SURVEY USED IN THE PREPARATION OF THESE DRAWINGS HAS NOT SPECIFIED A PERMANENT SURVEY MARK, VERTICAL AND HORIZONTAL HAVE BEEN ASSUMED TO BE ARBITRARY. THIS SURVEY HAS BEEN USED TO FORM THE BASIS OF THE DOCUMENTED DESIGN LEVELS.

EARTHWORKS

1. ALL EARTHWORKS ACTIVITIES AND MATERIALS ARE TO BE IN ACCORDANCE WITH AS3798 THESE CIVIL ENGINEERING DRAWINGS AND THE PROJECT SPECIFICATIONS. FILL BATTERS OUTSIDE PAVEMENTS ARE TO BE FILLED USING MATERIAL FROM THE ON-SITE CUT OPERATIONS, WHERE CUT MATERIAL IS TO BE FREE OF ORGANIC MATTER AND BUILDING WASTE.
2. ANY IMPORTED FILL MATERIAL OUTSIDE PAVEMENTS IS TO BE CBR15 MINIMUM, WELL GRADED AND FREE OF ORGANIC MATTER.
3. ALL BATTERS ARE TO BE 1 IN 6 UNLESS NOTED OTHERWISE.
4. INSTALL AND MAINTAIN EROSION AND SEDIMENT CONTROL SYSTEMS IN ACCORDANCE WITH THE LOCAL AUTHORITY'S REQUIREMENTS AND THESE CIVIL ENGINEERING DRAWINGS.
5. THE CONSTRUCTION SITE IS TO BE MAINTAINED TO MINIMIZE PONDING, EROSION AND DUST.
6. POSSIBLE CONTAMINATION OF THIS SITE HAS NOT BEEN INVESTIGATED. REPORT SUSPECT SOILS EXPOSED DURING SITE WORKS.
7. ANY RE-SPREAD TOPSOIL SHOULD BE SCREENED TO REMOVE ANY UNSUITABLE/OVERSIDED MATERIAL GREATER THAN 50mm. ALL UNSUITABLE MATERIAL FROM THE SCREENING PROCESS IS TO BE REMOVED FROM SITE.
8. THE PREPARATION OF THE EXISTING SUBGRADE AND EARTHWORKS ACTIVITIES ARE TO COMPLY WITH THE FOLLOWING:

9.1. PREPARE SUBGRADE AND CONSTRUCT EARTHWORKS UNDER ALL CONCRETE SLABS AND PAVEMENTS AND FOR A FURTHER 0.3m ALL ROUND.

9.2. STRIP ALL UNSUITABLE FOUNDATION MATERIAL (TOP SOIL, VEGETATION, ORGANICS, EXISTING UNCONTROLLED FILLING, ETC).

9.3. PROOF ROLL SUBGRADE USING A MINIMUM 10 TONNE LOADED WATER TRUCK OR APPROVED EQUIVALENT TO BOTH CUT AND FILL AREAS. REMOVE ANY FURTHER UNSUITABLE MATERIAL AND PROOF ROLL AGAIN.
- 9.4. FOR TENDER PURPOSES, IN ADDITION TO THE QUANTITIES ABOVE, 10m³ OF FURTHER UNSUITABLE MATERIAL TO BE EXCAVATED, REPLACED AND COMPACTED AS PER BELOW.

- 9.5. THE IMPORTED SUB-BASE BLENDED GRANULAR MATERIAL IS TO COMPLY WITH THE FOLLOWING:

9.5.1. THE MATERIAL IS TO CONTAIN SUFFICIENT FINES TO MINIMIZE MOISTURE PENETRATION.

9.5.2. THE MATERIAL IS TO BE CBR15 MINIMUM

9.5.3. THE SHRINKAGE PRODUCT OF THE MATERIAL (LINEAR SHRINKAGE x % PASSING 0.425mm SIEVE) IS TO BE IN THE RANGE OF 100 TO 350.

9.5.4. THE GRADING OF THE MATERIAL IS TO MEET GRADING A, B, C, OR D OF THE MAIN ROADS SPECIFICATION.

9.5.5. THE MATERIAL IS TO BE PLACED IN 200mm THICK LAYERS AND COMPACTED TO 95% STANDARD.
- 9.6. ALL EARTHWORKS MATERIAL (IMPORTED FILL AND CUT MATERIAL) OUTSIDE PAVEMENTS IS TO BE CBR15 MINIMUM, PLACED IN 200mm THICK MAXIMUM LAYERS AND COMPACTED TO 95% STANDARD COMPACTION.
- 9.7. PROOF ROLL THE COMPLETED EARTHWORKS. THE FINAL PASS SHALL BE TREATED AS TEST ROLLING IN ACCORDANCE WITH TESTING CLAUSE 5.5 OF AS3798.
10. LANDSCAPING AREAS ARE TO BE LIGHTLY TURNED. SPREAD AND PREPARE TOPSOIL TO ACCEPT LANDSCAPING INCLUDING TURF.

SIGNAGE AND LINE MARKING

1. ALL SIGNAGE AND LINE MARKING ACTIVITIES AND MATERIALS ARE TO BE IN ACCORDANCE WITH THESE CIVIL ENGINEERING DRAWINGS AND THE PROJECT SPECIFICATIONS.
2. ALL SIGNAGE AND LINE MARKING IS TO BE IN ACCORDANCE WITH THE MANUAL OF THE UNIFORM TRAFFIC CONTROL DEVICES (MUTCD) PREPARED BY THE DEPARTMENT OF TRANSPORT AND MAIN ROADS (TMR).
3. LINE MARKING IS TO COMPLY WITH THE FOLLOWING, UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS:

3.1. LINE MARKING IS TO BE WHITE UNLESS NOTED OTHERWISE.

3.2. SUFFICIENT TIME (3 DAYS MINIMUM) IS TO BE ALLOWED BETWEEN LAYING OF ASPHALT/PLACING CONCRETE AND LINE MARKING TO ALLOW FOR CURING OF ASPHALT/CONCRETE AND FOR CURING COMPONENTS TO DISSIPATE.

3.3. EDGE LINE MARKING IS TO BE 100mm WIDE.

3.4. LINE MARKING PAINT IS TO BE APPLIED BY A MECHANICAL SPRAYER.

3.5. THE FINISH OF THE PAVEMENT MARKING IS TO BE NON SLIP WITH SKID RESISTANCE LEVEL OF MINIMUM BRITISH PENDULUM NUMBER (BPN) 45 UNDER WET CONDITIONS.

3.6. PAINT IS TO HAVE A WET FILM THICKNESS BETWEEN 0.35mm AND 0.4mm.

3.7. THE REMOVAL OF UNREQUIRED LINE MARKING IS TO BE CARRIED OUT BY PAINT STRIPPER AND WATER BLASING OR APPROVED EQUIVALENT, AND IN ACCORDANCE WITH AUSTRROADS SPECIFICATIONS.
4. SIGNAGE IS TO COMPLY WITH THE FOLLOWING UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS:

4.1. SIGNAGE TO BE SIZE 'A' AS PER THE MUTCD.

4.2. SIGNAGE INSTALLATION HEIGHTS ARE TO BE AS PER DETAILS SHOWN ON CIVIL ENGINEERING DRAWINGS.

STORMWATER DRAINAGE

1. ALL STORMWATER DRAINAGE ACTIVITIES AND MATERIALS ARE TO BE IN ACCORDANCE WITH THE CIVIL ENGINEERING DRAWINGS AND THE PROJECT SPECIFICATIONS.
2. STORMWATER CULVERTS ARE TO COMPLY WITH THE FOLLOWING, UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS:

ELEMENT	GRADE	STANDARD
PRE-CAST RCBC - SMALL	-	AS1597.1

- 2.1. ALL BOX CULVERTS ARE TO BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATION AND REQUIREMENTS.
- 2.2. PLASTIC OR NATURAL RUBBER BANDAGE COVERS ARE TO BE PROVIDED BETWEEN CELL LENGTHS.
- 2.3. ANY LIFTING HOLES ARE TO BE PLUGGED FOLLOWING INSTALLATION WITH MANUFACTURER PLUGS OR DRY PACK MORTAR PRIOR TO BACKFILLING.
3. CULVERT LAYING IS TO COMMENCE AT THE DOWNSTREAM END OF EACH LINE AND TO CONTINUE WITH THE FEMALE PIPE ENDS FACING UPSTREAM.
4. CULVERT LAYING ACTIVITIES ARE TO COMPLY WITH THE FOLLOWING UNLESS NOTED OTHERWISE ON THE CIVIL ENGINEERS DRAWINGS:

4.1. DRAINAGE CULVERTS ARE TO BE LAID WITH A MINIMUM OF 100mm OF BEDDING MATERIAL TO THE BOTTOM, SIDES AND TOP OF PIPES.

4.2. SUBMIT DETAILS OF BEDDING MATERIAL FOR REVIEW BY THE CIVIL ENGINEER.

4.3. DRAINAGE CULVERT EXCAVATIONS ARE TO BE BACK FILLED WITH GRANULAR MATERIAL.

4.4. SUBMIT DETAILS OF BACKFILL MATERIAL FOR REVIEW BY THE CIVIL ENGINEER.

4.5. GRANULAR BACKFILL MATERIAL IS TO BE PLACED IN 150mm THICK LAYERS AND COMPACTED TO 95% STANDARD COMPACTION.
5. PRE-CAST HEADWALL END STRUCTURES ARE TO BE INSTALLED TO MANUFACTURER'S REQUIREMENTS UNLESS NOTED OTHERWISE.
6. ALL CONCRETE WORKS ARE TO BE IN ACCORDANCE WITH THE CONCRETE NOTES UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS.

EROSION AND SEDIMENT CONTROL

1. ALL EROSION AND SEDIMENT CONTROL IS TO BE IN ACCORDANCE WITH THE CMDG, QUEENSLAND STATE PLANNING POLICIES, IECA (INTERNATIONAL EROSION CONTROL ASSOCIATION AUSTRALASIA) AND THE DEPARTMENT OF ENVIRONMENT, TOURISM, SCIENCE AND INNOVATION REGULATIONS AND GUIDELINES.
2. ALL EROSION AND SEDIMENT CONTROL DEVICES AND TREATMENTS SHOWN ON THESE CIVIL ENGINEERING DRAWINGS ARE CONSIDERED BY THE CIVIL ENGINEER TO BE A MINIMUM STANDARD FOLLOWING COMPLETION OF THE BULK EARTHWORKS.
3. THE CONTRACTOR IS TO PREPARE SUITABLE EROSION AND SEDIMENT CONTROL PLANS FOR ALL STAGES OF CONSTRUCTION, TAKING INTO CONSIDERATION CONSTRUCTION STAGING AND MINIMISING THE AMOUNT OF "DISTURBED AREA" AT ANY GIVEN TIME. THE CONTRACTORS EROSION AND SEDIMENT CONTROL PLANS ARE TO BE PREPARED AND DISTRIBUTED TO THE CIVIL ENGINEER AND LOCAL AUTHORITY PRIOR TO THE PRE-START MEETING.
4. INSTALLATION OF ALL TEMPORARY EROSION AND SEDIMENT CONTROL MEASURES IS TO OCCUR PRIOR TO COMMENCEMENT OF EARTHWORKS OPERATIONS.
5. THE CONTRACTOR IS TO ENSURE ALL TEMPORARY EROSION AND SEDIMENT CONTROL MEASURES DOCUMENTED ON THESE CIVIL ENGINEERING DRAWINGS ARE MAINTAINED FOR THE DURATION OF THE CONSTRUCTION WORKS.
6. ALL SOIL EROSION AND SEDIMENT CONTROL MEASURES ARE TO BE CHECKED, MAINTAINED AND CORRECTED AS NECESSARY FOLLOWING RAIN EVENTS.
7. THE CONTRACTOR SHALL ENSURE DUST RESULTING FROM THE EARTHWORKS OPERATIONS IS KEPT TO A MINIMUM BY THE APPLICATION OF WATER TO THE EARTHWORKS AREA, ESPECIALLY ON DAYS OF WINDY WEATHER.
8. CONSTRUCTION VEHICLES ENTRY/EXIT SEDIMENT TRAPS ARE TO BE INSTALLED AT SITE ACCESS POINTS.
9. THE CONTRACTOR IS TO ENSURE ALL PERMANENT EROSION AND SEDIMENT CONTROL MEASURES AS SHOWN ON THESE CIVIL ENGINEERING DRAWINGS ARE INSTALLED PRIOR TO COMPLETION OF THE WORKS.
10. TURF IS TO BE SUPPLIED AND INSTALLED IN ACCORDANCE WITH THE LANDSCAPE ARCHITECTS SPECIFICATION.

11. THE CONTRACTOR IS TO MAINTAIN ALL PERMANENT EROSION AND SEDIMENT CONTROL DEVICES REGULARLY FOR A MINIMUM PERIOD OF 3 MONTHS OR UNTIL THE SATISFACTION OF THE CIVIL ENGINEER (I.E. TURF IS FIRMLY ROOTED).

QUALITY ASSURANCE TESTING

1. SUBGRADE

1.1. SOAKED CBR: 1 WITHIN VEHICLE PARKING EXTENTS.

1.2. FIELD DENSITY: 1 WITHIN VEHICLE PARKING EXTENTS.
2. SUB-BASE

2.1. PARTICLE SIZE DISTRIBUTION: 1 NO REQUIRED OF COMPACTED SAMPLE IF REQUESTED.

2.2. ATTERBERG LIMITS: 1 NO REQUIRED OF COMPACTED SAMPLE IF REQUESTED.

2.3. SOAKED CBR: 1 WITHIN VEHICLE PARKING EXTENTS.

2.4. FIELD DENSITY: 1 WITHIN VEHICLE PARKING EXTENTS.
3. STORMWATER DRAINAGE BACKFILL: 1 FIELD DENSITY TEST PER TRENCH BETWEEN PITS.
4. CONCRETE COMPRESSIVE STRENGTH TESTING: MINIMUM 1 TEST PER POUR (3 SAMPLES).
5. AS-CONSTRUCTED:

5.1. THE CONTRACTOR SHALL PROVIDE LEVELS AND DIMENSION INFORMATION SUITABLE TO CONFIRM TO THE SATISFACTION OF THE ENGINEER THAT THE WORKS HAVE BEEN CONSTRUCTED TO THE LEVELS AND DIMENSIONS SHOWN ON THE DRAWINGS. THE AS-CONSTRUCTED DATA PROVIDED MUST BE SUITABLE FOR THE PREPARATION OF AS DRAWINGS TO BE SUBMITTED TO THE CLIENT FOR APPROVAL. THE MINIMUM INFORMATION REQUIREMENTS ARE AS FOLLOWS:

5.1.1. VEHICLE PARKING HARDSTAND AND PATH EXTENTS AND LEVELS.

5.1.2. ALL DIMENSIONS SHALL BE PROVIDED IN METRES CORRECT TO 2 DECIMAL PLACES. ALL LEVELS SHALL BE ON AUSTRALIAN HEIGHT DATUM (AHD) AND THE AS-CONSTRUCTED SURVEY ON THE SAME COORDINATE SYSTEM AS THE DESIGN DOCUMENTATION SURVEY.
6. THE TOLERANCE REQUIREMENTS ON THE FINISHED SURFACE LEVEL OF CONCRETE HARDSTANDS AND PATHS SHALL BE AS FOLLOWS:

6.1. SUBGRADE SURFACE +0MM TO -25MM

6.2. PAVEMENT THICKNESS +20MM TO -10MM

6.3. FINISHED SURFACE HORIZONTAL ALIGNMENT ±10MM

6.4. FINISHED SURFACE VERTICAL/GEOMETRIC TOLERANCE ±10MM

6.5. FINISHED SURFACE PRIMARY TOLERANCE ±5MM

6.6. FINISHED SURFACE DEVIATION FROM 3M STRAIGHT EDGE ±5MM

6.7. FINISHED SURFACE CROSSFALL ±0.1%

6.8. FINISHED SURFACE RATE OF CHANGE OF CROSSFALL ±0.02% PER METRE

CONCRETE

1. ALL CONCRETE AND FORMWORK ACTIVITIES AND MATERIALS ARE TO BE IN ACCORDANCE WITH AS3600, AS3610, THESE CIVIL ENGINEERING DRAWINGS AND THE PROJECT SPECIFICATIONS.
2. ANY HOLES, RECESSES, CHASES AND CAST IN CONDUITS, PIPES OR OTHER ITEMS ARE TO BE APPROVED BY THE CIVIL ENGINEER, UNLESS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS.
3. REFER TO THESE CIVIL DRAWINGS FOR CONFIRMATION OF ALL LEVELS, STEPS AND FALLS.
4. CONSTRUCTION JOINTS ARE TO BE APPROVED BY THE CIVIL ENGINEER, UNLESS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS.
5. CONCRETE IS TO COMPLY WITH AS1379 AND THE FOLLOWING:

ELEMENT	GRADE	MAXIMUM AGGREGATE (mm)	SLUMP (mm)	SPECIAL REQUIREMENTS
PAVEMENT SLABS	S32	20	80	WATER/BINDER ≤ 0.46

6. UNLESS NOTED OTHERWISE IN THE PROJECT SPECIFICATIONS, SAMPLE AND TEST CONCRETE USING THE 'PROJECT ASSESSMENT' METHOD IN ACCORDANCE WITH AS1379. SUBMIT ALL TEST RESULTS TO THE CIVIL ENGINEER.
7. STEEL REINFORCING IS TO BE IN ACCORDANCE WITH THE STEEL REINFORCEMENT NOTES. PROVIDE STEEL REINFORCING IN THE LOCATIONS NOMINATED IN THESE CIVIL ENGINEERING DRAWINGS. THE POSITION OF STEEL REINFORCING IS TO BE MAINTAINED DURING CONCRETE PLACEMENT.
8. COVER TO STEEL REINFORCEMENT IS TO COMPLY WITH THE FOLLOWING:

ELEMENT	COVER (mm)	FACE
SLAB ON GROUND-EXTERNAL	45	TOP & EDGES
9. IF NOT DEFINED IN THE ARCHITECTURAL DRAWINGS OR THE PROJECT SPECIFICATIONS, FORMED CONCRETE SURFACES SHALL HAVE A SMOOTH, HARD, UNIFORM TEXTURE AND BE CLASS 3 FINISH TO AS3610.
10. IF NOT DEFINED IN THE ARCHITECTURAL DRAWINGS OR THE PROJECT SPECIFICATIONS, FINISH ALL EXTERNAL CONCRETE ELEMENTS WITH A LIGHT BROOM AFTER A SMOOTH STEEL TROWELLING.
11. CONCRETE ACTIVITIES ARE TO COMPLY WITH THE FOLLOWING, UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS:

11.1. PROVIDE 20mm CHAMFER TO ALL FORMED EDGES AND CORNERS OF CONCRETE ELEMENTS.

11.2. PROVIDE 20mm RADIUS TOOL TO ALL FINISHED EDGES OF CONCRETE ELEMENTS.

11.3. COMPACT ALL CONCRETE WITH THE USE OF IMMERSION VIBRATORS.

11.4. INSTALL SAW CUT JOINTS IN SLABS AS EARLY AS PRACTICABLE WITHOUT CAUSING UNACCEPTABLE DAMAGE TO JOINT EDGES.

11.5. THE FIRST POUR OF CONSTRUCTION JOINTS ARE TO BE FORMED AND THEN SCABBLED TO REMOVE LAITANCE AND EXPOSE AGGREGATE. AN APPROVED BONDING AGENT IS TO BE APPLIED IMMEDIATELY BEFORE THE SECOND POUR. CAST IN CONDUITS ARE NOT TO BE PLACED WITHIN THE COVER TO STEEL REINFORCING.

11.7. CAST IN CONDUITS ARE TO BE LOCATED AS FAR APART AS POSSIBLE AND A CLEAR SPACING OF TWICE THE LARGER CONDUIT DIAMETER IS TO BE MAINTAINED BETWEEN CONDUITS AND BETWEEN CONDUITS AND PARALLEL REINFORCING STEEL.
12. WET CONCRETE IS TO BE CURED BY THE APPLICATION OF AN EVAPORATION RETARDANT TO ALL HORIZONTAL CONCRETE SURFACES AS SOON AS PRACTICABLE AFTER LEVELLING OF THE WET CONCRETE. APPLY REPEAT APPLICATIONS AS REQUIRED TO MAINTAIN THE RETARDANT UNTIL FINISHING OPERATIONS COMMENCE.
13. HARDENED CONCRETE IS TO BE CURED CONTINUOUSLY FOR A MINIMUM OF 7 DAYS BY ONE OF THE FOLLOWING METHODS:

13.1. FULLY SEALING WITH PLASTIC SHEETING.

13.2. CONTINUOUS PONDING OF WATER.

13.3. APPLICATION OF A LIQUID MEMBRANE FORMING CURING COMPOUND IN ACCORDANCE WITH AS3799.
14. THE METHOD OF CURING IS TO BE SUBMITTED TO THE CIVIL ENGINEER. THE USE OF PROPRIETARY PRODUCT IS TO BE IN ACCORDANCE WITH MANUFACTURES SPECIFICATIONS.
15. ALL FORMWORK AND BACK PROPPING ACTIVITIES ARE TO BE IN ACCORDANCE WITH AS3610.

STEEL REINFORCEMENT

1. ALL STEEL REINFORCEMENT ACTIVITIES AND MATERIALS ARE TO BE IN ACCORDANCE WITH AS/NZS4671, AS3600, AS3700 AND AS3850.
2. THE CONTRACTOR IS TO PROVIDE EVIDENCE THAT ALL STEEL REINFORCEMENT IS CERTIFIED BY THE AUSTRALIAN CERTIFICATION AUTHORITY FOR REINFORCING AND STRUCTURAL STEELS (ACRS).
3. STEEL REINFORCING IS TO COMPLY WITH THE FOLLOWING:

STEEL REINFORCEMENT	DESIGNATION
N	DEFORMED RIBBED, 500MPa, NORMAL DUCTILITY BAR (D500N)
SL	DEFORMED RIBBED, 500MPa, SQUARELY CONFIGURED, LOW DUCTILITY MESH (D500SL)
R	ROUND, 250MPa, NORMAL DUCTILITY BAR (R250N)

4. LAPS TO STEEL REINFORCEMENT ARE TO COMPLY WITH THE FOLLOWING, UNLESS NOTED OTHERWISE IN THESE CIVIL ENGINEERING DRAWINGS:

STEEL REINFORCEMENT	LAP
N12	600mm
SLAB MESH	2 CROSS WIRES PLUS 25mm
5. STEEL REINFORCEMENT IS TO BE ONLY SPLICED AT THE LOCATIONS AND TO DETAILS SHOWN IN THESE CIVIL ENGINEERING DRAWINGS. ANY ADDITIONAL SPLICES ARE TO BE APPROVED BY THE ENGINEER BEFORE FABRICATION.
6. BEND STEEL REINFORCEMENT IN ACCORDANCE WITH AS3600.
7. DO NOT HEAT OR WELD STEEL REINFORCEMENT WITHOUT THE APPROVAL OF THE ENGINEER.
8. ALL STEEL REINFORCEMENT IS TO BE FABRICATED AND INSTALLED WITHIN THE TOLERANCES OF AS3600.
9. ALL STEEL REINFORCEMENT IS TO BE SUPPORTED IN ITS CORRECT POSITION BY PLASTIC BAR CHAIRS, PLASTIC TIPPED BAR CHAIRS OR CONCRETE SPACERS AT 800 MAX CRS IN BOTH DIRECTIONS.
10. PROVIDE 4 SETS OF TRIMMER BARS TO ALL PENETRATIONS. DO NOT CUT STEEL REINFORCEMENT TO CLEAR SMALL PENETRATIONS, BUT DISPLACE THE STEEL REINFORCEMENT TO CLEAR THE PENETRATION.

DRAWING LIST

C101	SPECIFICATION NOTES
C102	VEHICLE PARKING LAYOUT AND LINEMARKING PLAN AND DETAILS
C103	VEHICLE PARKING CONCRETE SLAB PLAN AND DETAILS
C104	VEHICLE PARKING EROSION AND SEDIMENT CONTROL PLAN AND DETAILS

Revision	Date	Revision Description	Drawn	Design	Checked	Client STANWELL	Project Janes and Stewart Structures Pty Ltd 120 William Street Po Box 1072 Rockhampton 4700 07 4922 1948 janes.and.stewart@jsstructures.com.au ABN 30 620 233 025		Project STANWELL NEW PROJECTS OFFICE	TENDER NOT FOR CONSTRUCTION PURPOSES	Project Number 25052	
1	03/11/2025	50% CD ISSUE	JC	JC	MD							Title C101
2	28/11/2025	90% CD ISSUE	JC	JC	MD							
3	19/12/2025	100% CD ISSUE	JC	JC	MD							
A	09/03/2026	FOR TENDER	JC	JC	MD							
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